MIL-L-13039C(AT)
23 June 1977
SUPERSEDING
MIL-L-13039B(AT)
12 May 1967

MILITARY SPECIFICATION

LADDERS, VEHICLE-BOARDING

This specification is approved for use by the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers portable, metal, step ladders for use by personnel in boarding vehicles.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS
MILITARY

MIL-E-52798	- Enamel, Alkyd, Camouflage.

STANDARDS
MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-193	- Painting Procedures, Tactical Vehicles (Tracked and Wheeled).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GSS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 2540

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(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction inspection. When specified, (see 6.2) a complete preproduction ladder sample(s) shall be furnished by the supplier. Unless otherwise specified (see 6.2), two samples shall be furnished. The approved samples and the preproduction inspection report shall be used as the production standard.

3.2 Design. The ladder shall be new. Each ladder shall be one complete unit conforming to the dimensions specified in figure 1. Total weight of the ladder shall not exceed 40 pounds.

3.3 Materials. The ladder shall be of all-metal construction as specified in figure 1. Materials not specifically designated shall be of such a quality as to meet the performance requirements. There is no requirement that virgin material be used.

3.4 Construction. The ladder shall be constructed as specified in figure 1. The ladder shall be complete so that when installed, it can be used for the operations specified herein.

3.4.1 Welding. All welding shall be performed in accordance with good commercial practice. In no event shall such processes, including peening and plugging, be used for reclaiming any part of the ladder without authorization of the procuring activity. The types of welds shall be in accordance with the symbols in figure 1. Warpage shall be limited to 1/4 inch out of plane.

3.4.2 Surfaces. All welded parts shall be cleaned and free of dirt, sand, scale or other harmful or extraneous materials. All burrs shall be removed and sharp edges and corners shall be rounded or beveled, except for those on the expanded metal treads.

3.5 Performance. Each step of the ladder shall be capable of withstanding, without damage or permanent set to any of the ladder components, a 500 pound load centrally applied on the top of the front step member perpendicular to the step.

3.6 Painting procedures. Except as otherwise specified, all conditioning, priming, and painting shall conform to MIL-STD-193.

3.6.1 Color. The ladder shall be finished in Forest Green per MIL-E-52798.

3.7 Identification. The ladder shall be marked for identification in accordance with MIL-STD-130.

3.8 Workmanship. Workmanship shall be of a quality to assure delivery of ladders which are free from defects resulting from defective material, improper manufacturing or assembly practices.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection of materials and components. The contractor is responsible for insuring that materials and components used were manufactured and inspected in accordance with requirements of referenced subsidiary specifications and standards to the extent specified herein, or if none, in accordance with the requirements of this specification.

4.2 Preproduction inspection. When specified, examination and testing of the preproduction sample(s) shall consist of examinations specified in 4.4.1 and 4.4.2, and the test specified in 4.6. Preproduction samples shall be as specified in 3.1. Failure of a preproduction sample to pass preproduction inspection shall be cause for rejection of the sample.

4.3 Lot. A lot shall consist of all ladders of the same materials and fabricating procedures offered for delivery at one time.

4.4 Sampling for visual and dimensional examination. Sampling for visual and dimensional examination shall be in accordance with MIL-STD-105, except as otherwise specified herein. The sample unit for the examinations shall be one completed fabricated and finished ladder. Inspection levels and acceptable quality levels (AQLs) shall be as specified in 4.4.3.

4.4.1 Visual examination. Visual examination shall be made in accordance with the following classification of defects.

Examine	Defects	Major	Minor
Design	Not as specified (see 3.2).	X	
Materials	Not type or kind specified (see 3.3).	X	
Construction	Not as specified (see 3.4)	X	
	Ladder not a complete unit (see 3.2).	X	
	Ladder weight exceeds specified (see 3.2).	X	
	Ladder not all metal construction (see 3.3).	X	
	Deformed or damaged components.	X	

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Examine	Defects	Major	Minor
Welding	Types of welds not as specified (see 3.4.1).	X	
	Warpage from welding exceeds specified limit (see 3.4.1).		X
	Surfaces of welded parts not free from scale or foreign matter (see 3.4.2).		X
	Sharp or burred edges not rounded or beveled, except for expanded metal treads (see 3.4.2).	X	
	Welds not sound, or are undercut (porosity, improper fusion or penetration).	X	
	Welds not free from slag inclusion, weld flash, spatter or flux.		X
	One spot weld loose or missing.		X
	More than one spot weld loose or missing.	X	
Painting procedures	Painting and finishing not as specified (see 3.6).		X
	Preparation and cleaning of surfaces prior to painting not as specified (see 3.6).		X
	Soft runs or sags in paint coat.	X	
	Floats or separation of color.	X	
	Area of no paint coat.		X
	Color not as specified (see 3.6.1).		X
Identification marking	Missing, illegible, incomplete, or not as specified.	X	

4.4.2 Dimensional examination. Examination shall be made for dimensions specified. Any dimension not within specified tolerances shall be classified as a defect.

4.4.3 Inspection levels and acceptable quality levels (AQLs) expressed in defects per 100 units shall be:

	Inspection Level	AQL	
		Major	Minor
For examination in 4.4.1	II	2.5	4.0
For examination in 4.4.2	S-2	(one class)	4.0

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4.5 Sampling for tests. From the samples which have passed the examinations specified in 4.4, samples shall be selected for tests specified in 4.6, in accordance with inspection level S-3 of MIL-STD-105, using the samples from the original lot as a basis for sampling for test. The samples selected shall be subjectd to the strength test specified in 4.6. The acceptable quality level (AQL) shall be 4.0 defects per 100 units.

4.6 Strength test. With the ladder in service position, a 500 pound load shall be applied on the top of the step member perpendicular to the step at an equal distance from each side rail of the ladder. The load shall be concentrated within the area of a 2 inch diameter. The test shall be applied separately to each step and the ladder observed for damage and checked for permanent deformation to determine compliance with 3.5.

4.7 Examination of preparation for delivery. An examination shall be made to determine that packaging, packing, and marking specified in the contract or order are complied with. The lot size shall be the number of shipping containers prepared for delivery. The sample unit shall be one container prepared for delivery. The inspection level shall be S-2 of MIL-STD-105, with an acceptable quality level (AOL) of 4.0 defects per 100 units.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with the packaging data sheet or packaging standard, as furnished at time of invitation for bids, for the level of preservation and packaging and level of packing specified (see 6.2).

6. NOTES

6.1 Intended use. The all metal ladders covered by this specification are utilized by personnel for boarding vehicles which will accompany them in the field. The ladders are portable and are carried on the vehicles.

6.2 Ordering data. Procurement documents should specify the following information:

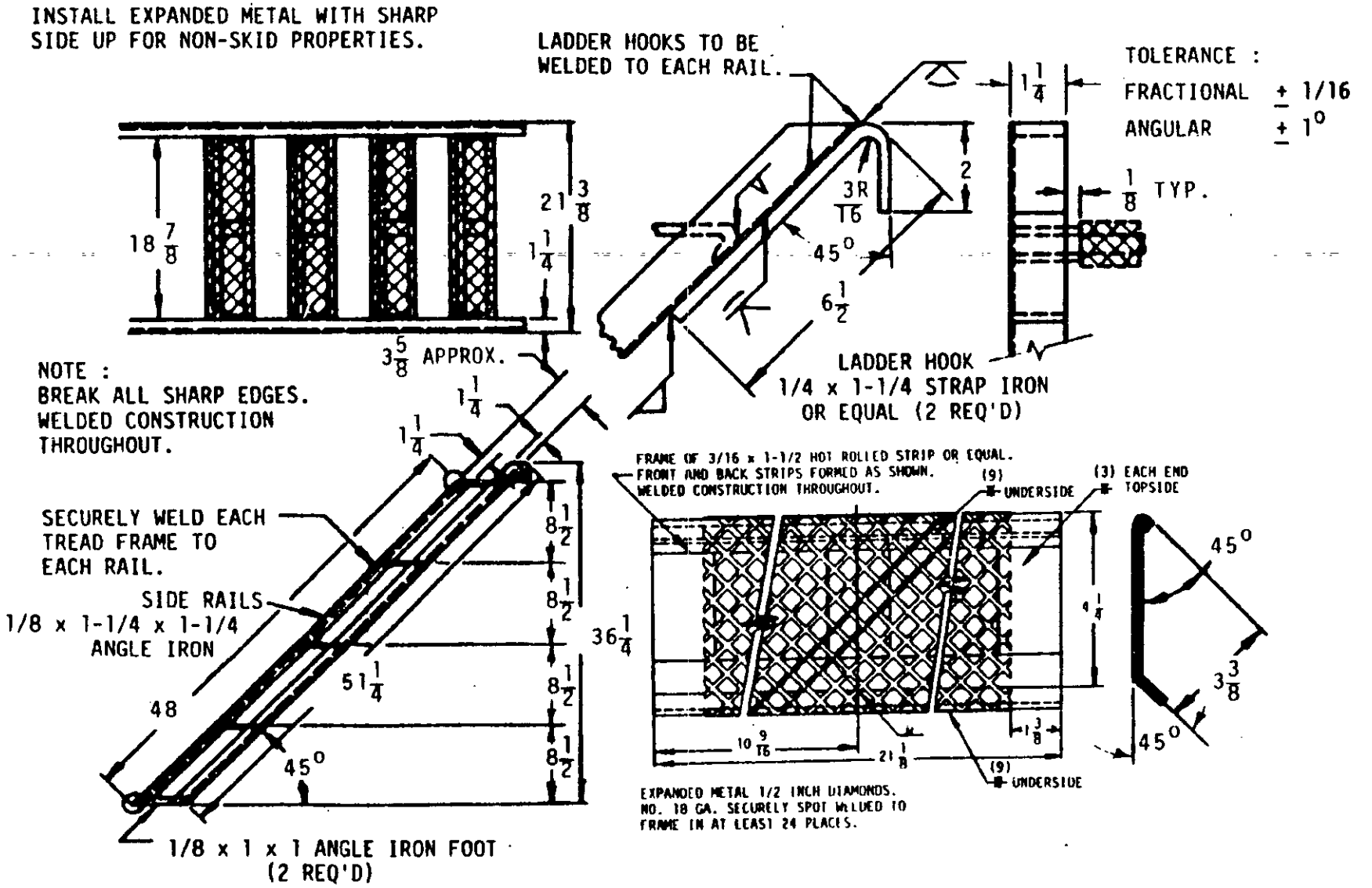
(a) Title, number and date of this specification.
(b) Preproduction inspection and quantity samples, if different (see 3.1).
(c) Selection of applicable level of preservation and packaging and level of packing (see 5.1).

Custodian:
Army - AT

Preparing activity:
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(Project No. 2540-A267)

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LADDER, STEP, STEEL

FIGURE 1

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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